

# Congenital Disorder of Glycosylation

## Type IIr (ATP6AP2-CDG / CDG2R)

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**Target disease:** Congenital Disorder of Glycosylation, Type IIr **MONDO:** MONDO:0026765 · **OMIM (phenotype):** 301045 · **Gene:** ATP6AP2 **Category:** Mendelian, X-linked recessive

**IMPORTANT IDENTIFICATION NOTE.** The MONDO term MONDO:0026765 ("congenital disorder of glycosylation, type IIr", synonym **CDG2R**) is cross-referenced (MONDO:equivalentTo) to **OMIM 301045**, **MedGen C5393313 (UID 1717186)**, **GARD 0025491**, **UMLS C5393313**, and **DOID:0051048**. This is the **X-linked ATP6AP2-CDG** described by Rujano et al. 2017 ([P 29127204](#)).

It is a **different disease** from CCDC115-CDG, which is CDG type **IIq** (OMIM 616828, autosomal recessive). This report covers ATP6AP2-CDG (the correct disease for MONDO:0026765). CCDC115-CDG and other V-ATPase-assembly CDGs (TMEM199-CDG/CDG-IIp, ATP6AP1-CDG, VMA21-CDG) are discussed only as differential/comparators.

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## 1. Disease Information

**Overview.** ATP6AP2-CDG (CDG-IIr / CDG2R) is an ultra-rare, X-linked recessive multisystem congenital disorder of glycosylation caused by hypomorphic missense variants in *ATP6AP2*, which encodes an accessory subunit of the vacuolar-type H<sup>+</sup>-ATPase (V-ATPase), also known as the (pro)renin receptor. Affected males present in infancy with liver disease (hepatosteatorrhea progressing to cirrhosis/liver failure), immunodeficiency (hypogammaglobulinemia with recurrent infections), cutis laxa, and mild psychomotor/intellectual impairment, together with a combined N-/O-linked (Golgi/type 2) glycosylation defect.

**Key identifiers.** - **MONDO:** MONDO:0026765 - **OMIM phenotype:** 301045 (CDG2R); **OMIM gene:** 300556 (ATP6AP2) - **MedGen:** C5393313 / UID 1717186 · **GARD:** 0025491 · **UMLS:** C5393313 · **DOID:** 0051048 - **Orphanet:** Falls under the broad group "Congenital disorder of

glycosylation" (ORPHA:137); a dedicated ATP6AP2-CDG ORPHA code is not consistently assigned (data not clearly available). - **ICD-11:** 5C51.2 (Congenital disorders of glycosylation); **ICD-10:** E77.8. **MeSH:** Congenital Disorders of Glycosylation (D018981).

**Synonyms / alternative names:** ATP6AP2-CDG; CDG2R; CDG-IIr; Congenital disorder of glycosylation type 2R; ATP6AP2 congenital disorder of glycosylation; (in older literature grouped with "V-ATPase CDGs" and "autophagic liver disease").

**Data source type:** Disease-level knowledge aggregated from a small number of **individual patient case reports/series** (fewer than ~7 reported individuals with the CDG phenotype as of 2025), plus functional studies in cells and model organisms. No EHR/registry-scale data exist.

## 2. Etiology

**Primary cause — genetic.** Hemizygous (males) germline **loss-of-function/hypomorphic missense variants in ATP6AP2** (Xp11.4). The disease is monogenic and Mendelian; there is no environmental or infectious cause of the disorder itself (though the resulting immunodeficiency predisposes to infections — a *consequence*, not a cause).

**Genetic risk factors / causal variants (human clinical):** - p.Arg71His (R71H) and p.Leu98Ser (L98S) — the two original missense variants (Rujano 2017, ).

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- c.185G>A p.Gly62Glu (G62E) — third reported missense (Fang 2023, ).

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- All lie in the **extracellular/luminal domain** of ATP6AP2; all are hypomorphic.

**Environmental risk factors:** None established. Being a monogenic X-linked disorder, the dominant "risk factor" is **male sex** (hemizyosity) and carrier-mother inheritance.

**Protective factors:** In females, the second (wild-type) *ATP6AP2* allele and X-inactivation generally protect heterozygous carriers (though a mildly affected female with a *splicing* variant has been reported — Raynor 2025, ).

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No dietary/environmental protective factors are known.

**Gene–environment interactions:** Not characterized. Because ATP6AP2 controls lysosomal acidification/autophagy and lipid handling, nutritional/metabolic state may modulate the steatotic phenotype (inferred from *Drosophila* lipid-metabolism findings,

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but this is not clinically demonstrated.

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### 3. Phenotypes

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Onset is **congenital/infantile**; the course is **severe and progressive**, dominated by liver disease. Frequencies are qualitative given the tiny cohort.

| Phenotype                                              | Type                      | HPO term                   | Onset / severity / frequency |
|--------------------------------------------------------|---------------------------|----------------------------|------------------------------|
| Hepatic steatosis / fatty liver                        | Lab/imaging/<br>clinical  | HP:0001397                 | Infantile;<br>consistent     |
| Liver failure                                          | Clinical sign             | HP:0001399                 | Infantile; severe;<br>common |
| Cirrhosis                                              | Clinical/<br>pathology    | HP:0001394                 | Progressive                  |
| Elevated transaminases                                 | Lab<br>abnormality        | HP:0002910                 | Consistent                   |
| Jaundice / cholestasis                                 | Clinical sign             | HP:0000952 /<br>HP:0001396 | Reported                     |
| Coagulopathy                                           | Lab<br>abnormality        | HP:0001928                 | Reported (Fang<br>2023)      |
| Hypogammaglobulinemia                                  | Lab<br>abnormality        | HP:0004313                 | Common                       |
| Recurrent infections<br>(immunodeficiency)             | Clinical                  | HP:0002719                 | Common                       |
| Cutis laxa                                             | Physical<br>manifestation | HP:0000973                 | Characteristic               |
| Global developmental delay /<br>psychomotor impairment | Behavioral/<br>neuro      | HP:0001263                 | Mild, variable               |
| Intellectual disability, mild                          | Behavioral                | HP:0001256                 | Mild                         |
| Dysmorphic facial features                             | Physical                  | HP:0001999                 | Some patients                |
| Growth retardation                                     | Clinical                  | HP:0001510                 | Reported                     |
| Anemia                                                 | Lab<br>abnormality        | HP:0001903                 | Reported                     |
| Cardiomegaly                                           | Clinical/imaging          | HP:0001640                 | Reported (Fang<br>2023)      |
|                                                        | Lab biomarker             |                            |                              |

| Phenotype                                           | Type | HPO term                               | Onset / severity / frequency |
|-----------------------------------------------------|------|----------------------------------------|------------------------------|
| Abnormal transferrin glycosylation (type 2 pattern) |      | HP:0012345<br>(abnormal glycosylation) | Diagnostic hallmark          |

**Quality-of-life impact:** Severe. Liver failure requiring transplantation, chronic immunodeficiency requiring immunoglobulin support, and developmental impairment substantially affect survival and daily functioning. Formal QoL instruments (EQ-5D/SF-36) have not been applied to this ultra-rare disease.

**Evidence quotes:** MedGen definition (per Rujano 2017): *"an X-linked recessive disorder characterized by infantile onset of liver failure, recurrent infections due to hypogammaglobulinemia, and cutis laxa. Some patients may also have mild intellectual impairment and dysmorphic features. Laboratory studies showed defective glycosylation of serum transferrin in a type 2 pattern."*

## 4. Genetic / Molecular Information

- **Causal gene:** *ATP6AP2* (ATPase H<sup>+</sup> transporting accessory protein 2; a.k.a. (pro)renin receptor, PRR, ATP6IP2). **HGNC:18305**, **NCBI Gene 10159**, **Ensembl ENSG00000182220**, gene **OMIM 300556**, cytoband **Xp11.4** (GRCh37 chrX:40,440,146–40,465,889). RefSeq NM\_005765. **UniProt O75787**.

- **Pathogenic variants (all germline, hemizygous, missense):**

- p.Arg71His (R71H) — Rujano 2017

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- p.Leu98Ser (L98S) — Rujano 2017

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- c.185G>A p.Gly62Glu (G62E) — Fang 2023

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- All in the **luminal/extracellular domain**; **loss-of-function/hypomorphic** consequence.

- **Variant classification (ACMG/AMP):** the recurrent CDG missense variants are reported pathogenic/likely pathogenic in the primary literature; most other *ATP6AP2* variants in ClinVar are VUS or relate to the neurological allelic disorders.
- **Allele frequency:** Effectively absent from population databases (gnomAD) — consistent with a severe, ultra-rare X-linked disease.
- **Somatic vs germline:** Germline only.
- **Allelic disorders (same gene, different variant class/phenotype):**
- *ATP6AP2* **splicing** variants → X-linked intellectual disability (Hedera type) and X-linked parkinsonism with spasticity (OMIM 300423 / 300911), acting via **exon-4 skipping ( $\Delta e4$ )** and **~50% reduced full-length protein** (Edelman 2022,

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Raynor 2025,

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- **Modifier genes:** None identified. **Epigenetic changes:** X-inactivation determines expression in carrier females; no disease-specific methylation signature reported.
- **Chromosomal abnormalities:** Not applicable (point mutations); the locus is Xp11.4.

Quote

(Raynor

2025,

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*"ATP6AP2 missense variants lead to hepatopathy, immunological abnormalities, cutis laxa and only mild intellectual disability with N-/O-glycosylation defects (ATP6AP2-CDG; OMIM#301045)."*

## 5. Environmental Information

Not applicable as a cause. No toxins, radiation, lifestyle, or infectious agents cause ATP6AP2-CDG. Infections occur **secondary** to the intrinsic humoral immunodeficiency (hypogammaglobulinemia). CHEBI-relevant chemical entities implicated downstream include cholesterol (CHEBI:16113) and triglycerides/lipids accumulating in liver.

## 6. Mechanism / Pathophysiology

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### Ordered causal chain (initiating lesion → clinical manifestation):

1. A hemizygous **hypomorphic missense variant in the *ATP6AP2* extracellular/luminal domain** (R71H, L98S, G62E) **reduces functional *ATP6AP2***, an ER-localized accessory subunit of the V-ATPase.
2. → **Impairs assembly/biogenesis of the V-ATPase proton pore (V0 sector) in the endoplasmic reticulum** (*ATP6AP2* interacts with V0-assembly components; its ER localization is crucial for V-ATPase activity).
3. → **Reduces V-ATPase-dependent acidification of the endolysosomal system and Golgi.**
4. Branch A (**glycosylation**): abnormal Golgi luminal pH → **dysfunctional/mislocalized glycosyltransferases** → **combined N-/O-linked hypoglycosylation** (serum transferrin type 2 pattern) → cutis laxa (defective ELN/elastic-fiber glycoprotein processing, inferred) and multisystem glycoprotein dysfunction, including **hypogammaglobulinemia/immunodeficiency**.
5. Branch B (**autophagy/metabolism**): defective **lysosomal acidification** → **reduced mTOR signaling** and **dysregulated (impaired) autophagy** → failure of hepatic lipid turnover/lipophagy → **hepatosteatosis** → hepatocyte injury, **fibrosis, cirrhosis, and liver failure**; altered lipid metabolism (demonstrated in *Drosophila* fat body and by metabolomics in cell models).
6. → Clinical convergence: **infantile liver failure, immunodeficiency, cutis laxa, mild psychomotor/intellectual impairment**.

*Demonstrated vs inferred:* Steps 1–3 and 5 are experimentally supported (mouse liver, *Drosophila* fat body, patient cells). The precise Golgi glycosyltransferase step (4) is inferred by analogy to other V-ATPase/Golgi-pH CDGs; cutis-laxa causation is mechanistically inferred.

**Molecular pathways:** V-ATPase-dependent organelle acidification; **mTOR (MTORC1) signaling** (KEGG hsa04150); **macroautophagy/lysosomal degradation** (KEGG hsa04140); lipid metabolism. **Cellular processes / GO:** vacuolar acidification (GO:0007035), regulation of macroautophagy (GO:0016241), TORC1 signaling (GO:0038202), protein N-/O-glycosylation (GO:0006487 / GO:0006493), lysosomal lumen acidification (GO:0007042). **Protein dysfunction:** hypomorphic loss of function of an assembly accessory subunit (not aggregation). **Immune involvement:** humoral immunodeficiency (hypogammaglobulinemia). **Tissue damage:** hepatic steatosis → steatohepatitis-like injury → fibrosis/cirrhosis.

**Molecular profiling:** Untargeted metabolomics of G62E-expressing cells showed downregulation of lipid-metabolism metabolites (Fang 2023, ).

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Serum glycoproteomics: hypoglycosylated transferrin (type 2 CDG pattern) and abnormal apolipoprotein/O-glycan markers.

**Quote** (Cannata Serio 2018, ):

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*"these phenotypes are the result of a pathogenetic cascade that includes impaired V-ATPase assembly, defective lysosomal acidification, reduced MTOR signaling and autophagic misregulation."*

**Cell types (CL) / compartments (GO CC):** hepatocyte (CL:0000182), B cell / plasma cell (CL:0000236 / CL:0000786), fibroblast (CL:0000057); endoplasmic reticulum (GO:0005783), lysosome (GO:0005764), Golgi apparatus (GO:0005794), vacuolar proton-transporting V-ATPase complex (GO:0016471).

## 7. Anatomical Structures Affected

- **Primary organ: Liver** (UBERON:0002107) — steatosis, cirrhosis, failure.
- **Secondary/other systems: Immune system** (UBERON:0002405; hypogammaglobulinemia), **skin/connective tissue** (UBERON:0002097; cutis laxa), **central nervous system/brain** (UBERON:0000955; mild developmental/intellectual impairment), and occasionally **heart** (UBERON:0000948; cardiomegaly).
- **Body systems:** digestive/hepatobiliary, immune, integumentary/connective, nervous, (cardiovascular).
- **Tissue/cell level:** hepatic parenchyma (hepatocytes), antibody-producing B/plasma cells, dermal elastic connective tissue/fibroblasts.
- **Subcellular:** ER, lysosome, Golgi (V-ATPase complex) — see GO terms above.
- **Localization/lateralization:** systemic/bilateral (not lateralized).

## 8. Temporal Development

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- **Onset:** Congenital defect; clinically **infantile** (liver disease often within the first months to first year of life; e.g., presentation at 11 months in Fang 2023).
  - **Onset pattern:** subacute–chronic hepatic deterioration; recurrent infections from infancy.
  - **Progression:** progressive hepatic disease (steatosis → cirrhosis → failure); developmental impairment relatively stable/mild.
  - **Duration:** chronic, lifelong; potentially life-limiting without liver transplantation.
  - **Critical period / intervention window:** early infancy — timely diagnosis enables immunoglobulin replacement and consideration of liver transplantation before decompensation.
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## 9. Inheritance and Population

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- **Inheritance: X-linked recessive.** Affected individuals are almost exclusively **hemizygous males**; heterozygous females are usually unaffected (a mildly affected female with a *splicing* variant is reported,

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- **Penetrance:** high/complete in hemizygous males for reported pathogenic missense; **expressivity variable** (hepatic severity and neuro-involvement differ).
  - **Epidemiology: Ultra-rare.** Only a handful of CDG-phenotype patients reported worldwide ( $\approx 3$  males in the original two families plus subsequent individual case reports). Prevalence/incidence are not quantifiable ( $\ll 1/1,000,000$ ); no registry data.
  - **Carrier frequency:** Not established; variants essentially absent from gnomAD.
  - **Founder effect / consanguinity:** None reported (X-linked, transmitted through carrier mothers).
  - **Sex ratio:** Strongly male-predominant (X-linked recessive).
  - **Geographic distribution:** Reported cases from Europe (Germany/Portugal) and China — no geographic clustering; reflects sporadic ascertainment.
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## 10. Diagnostics

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- **First-line biomarker: Serum transferrin isoform analysis** (isoelectric focusing or capillary electrophoresis / mass spectrometry) showing a **CDG type 2 pattern** (loss of sialic acid/galactose from a Golgi defect). Combined with **abnormal O-glycosylation** markers (e.g., apolipoprotein C-III).
  - **Laboratory tests:** elevated aminotransferases (ALT/AST), cholestatic markers/bilirubin, **low serum immunoglobulins (IgG)**, coagulopathy (reduced clotting factors), hypercholesterolemia/dyslipidemia, anemia.
  - **Imaging:** hepatic ultrasound/MRI showing steatosis/hepatomegaly/cirrhosis; echocardiography if cardiomegaly.
  - **Histopathology (liver biopsy):** steatosis, fibrosis/cirrhosis, autophagic/lysosomal abnormalities; immunohistochemistry for ATP6AP2.
  - **Genetic testing (definitive):** WES/WGS or targeted **CDG/hepatopathy gene panels** including *ATP6AP2*; single-gene sequencing confirms hemizygous missense. **RNA-seq of patient fibroblasts** is valuable to characterize splicing variants (relevant for the allelic neurological disorders). CMA/karyotype/FISH not indicated for point mutations.
  - **Functional confirmation:** demonstration of hypoglycosylation and, in research settings, V-ATPase/lysosomal-acidification and mTOR/autophagy assays; complementation studies.
  - **Differential diagnosis: Wilson disease**, mitochondrial hepatopathies, Niemann-Pick C, other **V-ATPase-assembly CDGs** (ATP6AP1-CDG, TMEM199-CDG/CDG-IIp, CCDC115-CDG/CDG-IIq, VMA21-CDG), ARC syndrome and other cutis-laxa CDGs (ATP6V0A2-CDG), and primary immunodeficiencies with liver disease. Distinguishing feature: X-linked inheritance + CDG type 2 transferrin + the triad liver/immune/cutis laxa.
  - **Screening:** No newborn-screening test (CDG type 2 is not captured by standard NBS). **Carrier testing** for at-risk female relatives and **cascade/prenatal testing** once the familial variant is known.
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## 11. Outcome / Prognosis

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- **Severity/mortality:** Serious; **liver failure is the principal life-threatening complication**. Death from hepatic failure has occurred in related V-ATPase CDGs; ATP6AP2-CDG patients have required **liver transplantation** (e.g., Fang 2023).

- **Life expectancy:** Reduced without treatment; can be extended by transplantation and immunoglobulin support. Precise survival statistics are unavailable (too few patients).
- **Morbidity/disability:** chronic liver disease, recurrent infections, developmental/intellectual impairment, connective-tissue (cutis laxa) manifestations.
- **Complications:** cirrhosis, portal hypertension, coagulopathy, infections, growth failure, cardiomegaly.
- **Recovery potential:** Liver transplantation addresses the hepatic phenotype but **does not correct the systemic glycosylation defect, immunodeficiency, or neurodevelopmental features.**
- **Prognostic factors:** age at diagnosis, hepatic reserve at presentation, timeliness of transplant, severity of immunodeficiency. **Prognostic biomarker:** degree of transferrin hypoglycosylation and hepatic synthetic function (albumin/coagulation).

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## 12. Treatment

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**No disease-specific/curative pharmacotherapy exists.** Management is **supportive and organ-directed** (NCIT: Supportive Care, C15277):

- **Hepatic:** management of chronic liver disease and, for end-stage disease, **orthotopic liver transplantation** (NCIT: Liver Transplantation, C15311) — performed in reported patients.
- **Immunologic: immunoglobulin (IgG) replacement therapy** (IVIG/SCIG) and infection prophylaxis/prompt treatment for hypogammaglobulinemia (NCIT: Immunoglobulin Therapy, C158353; Intravenous Immunoglobulin Therapy).
- **Nutritional/metabolic:** management of coagulopathy (vitamin K, factor support), dyslipidemia, and growth/nutrition.
- **Neurodevelopmental:** early intervention, physical/occupational/speech therapy for developmental delay.
- **Cardiac/other:** monitoring and symptomatic management as needed.

**Pharmacogenomics:** Not established. **Advanced therapeutics (gene/cell/RNA therapy, targeted therapy):** none approved or in trials; **no ClinicalTrials.gov entries specific to ATP6AP2-CDG** (as of 2025). Given the mechanism, mTOR/autophagy modulation is a

conceptual research target (not clinically validated). **Treatment response/adverse events:** limited to individual case experience; transplantation carries standard surgical/immunosuppression risks.

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## 13. Prevention

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- **Primary prevention:** Not possible for a monogenic disorder; **genetic counseling** for carrier families (X-linked recurrence risk: carrier mother → 50% of sons affected, 50% of daughters carriers).
  - **Secondary prevention:** early diagnosis via CDG screening in infants with unexplained liver disease + immunodeficiency + cutis laxa; **cascade genetic testing** of relatives.
  - **Reproductive options:** **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** once the familial *ATP6AP2* variant is known; carrier testing of at-risk females.
  - **Tertiary prevention:** immunoglobulin replacement to prevent infections; hepatology surveillance to time transplantation; developmental support.
  - **Immunization / public-health / environmental measures:** standard vaccinations as tolerated within immunodeficiency management; no population-level or environmental prevention applies.
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## 14. Other Species / Natural Disease

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- **Orthologs (comparative/experimental, not natural disease):** mouse *Atp6ap2* (NCBI Gene 70495; NCBI Taxon 10090), rat *Atp6ap2*, zebrafish *atp6ap2*, *Drosophila melanogaster* ortholog (VhaPRR; Taxon 7227). ATP6AP2/V-ATPase function is **highly evolutionarily conserved** (yeast to human).
  - **Natural disease in other species: None reported.** No naturally occurring ATP6AP2-CDG in companion animals/wildlife (OMIA: not listed). No breed-specific (VBO) associations.
  - **Zoonotic potential / transmission:** Not applicable (genetic disease).
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## 15. Model Organisms

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- **Mouse (mammalian):** Liver-directed **Atp6ap2 deficiency** causes **hypoglycosylation of serum proteins and autophagy defects**, recapitulating hepatic/glycosylation features (Rujano 2017,

[P 29127204](#)

*Atp6ap2* is essential; complete knockout is lethal, so conditional/tissue-specific approaches are used. Resource: MGI.

- **Drosophila (invertebrate):** Introduction of a human missense mutation caused **reduced survival and altered lipid metabolism**, with **defective lysosomal acidification and reduced mTOR signaling** and autophagic dysregulation in the liver-like **fat body**

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[P 29127204](#)

Resource: FlyBase.

- **Cell / in vitro models:** **HEK293T** cells expressing p.Gly62Glu show dysregulated autophagy/ mTOR and altered lipid metabolites (Fang 2023,

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**patient fibroblasts** (glycosylation + splicing/RNA-seq studies); **iPSC-derived neural progenitor cells** used for allelic splicing variants (Edelman 2022,

[P 35779466](#)

- **Phenotype recapitulation:** Models reproduce the **glycosylation, lysosomal-acidification, mTOR/autophagy, and lipid/steatosis** phenotypes well. **Limitations:** they incompletely capture the **immunodeficiency** and **cutis laxa** components of the human disease.
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## Evidence Summary (Key PMIDs)

| PMID                                             | Type                              | Contribution                                                                                        |
|--------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------|
| 29127204 (Rujano 2017, <i>J Exp Med</i> )        | Human + mouse + <i>Drosophila</i> | Landmark: identifies <i>ATP6AP2</i> missense (R71H, L98S) as cause; defines phenotype and mechanism |
| 29388887 (Cannata Serio 2018, <i>Autophagy</i> ) | Review/commentary                 | Summarizes pathogenetic cascade (V-ATPase assembly → lysosomal acidification → mTOR → autophagy)    |
| 38075676 (Fang 2023, <i>Front Genet</i> )        | Human case + cell model           | New variant G62E; liver transplant; metabolomics of lipid pathway                                   |
| 41131679 (Raynor 2025)                           | Human                             | Missense-vs-splicing genotype–phenotype; first affected female; glycosylation in splicing variants  |
| 35779466 (Edelman 2022)                          | Human + iPSC                      | Reduced-gene-dosage ( $\Delta e4$ ) mechanism for splicing/neurological allelic disorders           |

**Evidence-source classification:** predominantly **human clinical case reports** supported by **model-organism (mouse, *Drosophila*)** and **in vitro/cellular** functional studies; no computational-only claims relied upon.

## Limitations and Future Directions

- **Ultra-rare disease:** conclusions rest on <~7 CDG-phenotype patients; frequencies, prevalence, penetrance nuances, and survival statistics are imprecise.
- **Ontology/registry gaps:** a dedicated Orphanet code and standardized HPO-frequency annotations are limited.
- **Mechanistic gaps:** the exact link from V-ATPase/pH defect to cutis laxa and to B-cell/immunoglobulin failure is inferred, not fully proven; the puzzle of why missense→hepatic/immune while splicing→neurological remains open.
- **Therapeutics:** no targeted therapy; mTOR/autophagy modulation and gene-directed approaches are unexplored research avenues.

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